

# ABHINAV KAUSHIK

Senior Data Scientist

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## SUMMARY

Senior Data Scientist with 8+ years turning large, imperfect datasets into reliable methods and measurable results across Healthcare, Telecom and AIOps, with an MSc in Machine Learning and AI. Builds Python, Pandas and PySpark data processing pipelines, statistical models for prediction, trend analysis and anomaly detection, and clear dashboards for stakeholders. Also applies GenAI and agentic workflows to automate data processing, taking prototypes to production with Docker and MLflow. Alongside this, I am building ThinkTree.AI, a non-profit AI learning platform used by NGO teachers to support students with ADHD.

## PROFESSIONAL EXPERIENCE

**Senior Data Scientist** | United Health Group Bengaluru, India | Nov 2025 - Present

- Deployed a production lab-extraction agent with Claude Opus 4.7 and Amazon Textract, integrated with LIMS, handling about 1,800 lab documents and automating the work behind 800 Jira tickets and PRD tasks.
- Engineered a production multi-agent analytics chatbot in LangGraph for AARP medical plans, answering business questions with PySpark analysis, multi-turn reasoning and live charts, cutting manual analysis by 98%.
- Built automated evaluation and observability pipelines for Question Answering and Summary Agents with LLM evaluators, tracing and failure detection, reducing human-in-the-loop review by 43%.
- Fine-tuned a cross-encoder on positive and negative pairs to re-rank dense retrieval over AARP survey data, improving retrieval accuracy.

**Senior Machine Learning Engineer** | Viavi Solutions Chennai, India | Sep 2024 - Nov 2025

- Developed an internal RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate against existing organizational solutions.
- Built an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN to trace the cascading effect of network alarm floods back to the root event, improving semantic alarm classification by 73.5%.
- Prototyped 15+ AI applications for AIOps and GeoPerformance clients from client requirements, with 7 moving into production or paid client work.

**Data Scientist** | Viavi Solutions Bengaluru, India | Nov 2021 - Sep 2024

- Deployed a production SLA breach prediction system with LightGBM, FastAPI and MLflow, using GPT-3.5 for real-time root-cause analysis and remediation recommendations per breach, reducing downtime by 67%.
- Analysed trends and anomalies across 15+ KPIs on 500+ telecom network devices with Holt-Winters demand forecasting and Isolation Forest, enabling earlier detection and proactive network planning.
- Delivered technical POCs and consulting for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

**Data Analyst** | Optum Gurugram, India | Jun 2021 - Nov 2021

- Implemented a reranker for an existing recommendation system serving individual and child plan recommendations, optimizing renewal suggestions and increasing conversion by 2%.
- Automated data integration with a Python and Flask parser that extracted and transformed raw data into structured datasets, feeding downstream analytics and reporting.

**Data Engineer** | Nippon Data Systems Gurugram, India | Nov 2019 - May 2021

- Automated BAU medical-audit reporting with scalable SQL and PySpark data processing pipelines on large healthcare datasets, cutting manual reporting effort by 95%.
- Analysed and aggregated large data sources with complex SQL using multi-table joins, CTEs and window functions, delivering ad-hoc reports and analyses to multiple stakeholder teams.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

**Associate Consultant** | Saksoft Gurugram, India | Dec 2017 - Nov 2019

- Developed an XGBoost propensity model predicting personal-loan applications within 7 days of contact, enabling targeted outreach and customer prioritization.
- Designed clear data visualizations and customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

## PROJECTS

**Think Tree** | ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

**Retail Sales Trend Estimation**

- Built a trend-estimation pipeline for retail sales panels with incomplete store coverage, reweighting samples and imputing missing outlets with gradient boosting in PySpark and Pandas.

- Validated estimates against held-out stores with unit-tested checks and bias diagnostics, then packaged the pipeline as a command-line tool in Docker for scheduled runs on Azure.
- Technologies:** Python, Apache Spark, Pandas, NumPy, Scikit-learn, LightGBM, Docker, Azure

**RFP Multi-Agent Proposal Automation Platform**

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

**Financial Fraud Detection** | Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

**Restaurant Feedback Analyzer** | Aspect-Based Sentiment Analysis

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

**EDUCATION**

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|--------------------------------------------------------------------------------------|---------------------|
| <b>Master of Science, Machine Learning and AI</b>   Liverpool John Moores University | Jan 2021 - Jan 2023 |
| <b>Bachelor of Technology, Information Technology</b>   Jamia Hamdard                | Aug 2013 - Aug 2017 |

**TECHNICAL SKILLS**

**AI / GenAI:** Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract  
**Machine Learning:** PyTorch, Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning  
**Evals / Observability:** LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis  
**Programming / Data:** Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix  
**MLOps / Cloud:** MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, GCP, Azure, Vercel  
**Data Platforms:** Databricks, Databricks SQL, Apache Spark, Supabase  
**Databases:** TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB  
**Search:** Apache Solr, Lucene Indexing, Information Retrieval  
**BI:** Power BI, QlikSense, SAP HANA

**LANGUAGES**

English: Native

**ADDITIONAL INFORMATION**

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies